

GIS Mapping of Species Distribution

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```
tinytex::install_tinytex() rmarkdown::render("Navarroza_gis.rmd", output_format = "pdf_document")
```

```
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union
```

```
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(terra)
```

```
## terra 1.9.34
```

```
library(ggplot2)  
setwd("C:/Users/Admin/downloads")  
# 1.  
frogs_raw <- read.csv("Frog.csv", sep = "\t")  
frogs_clean <- frogs_raw %>%  
  filter(!is.na(decimalLatitude), !is.na(decimalLongitude)) %>%  
  distinct(occurrenceID, .keep_all = TRUE) %>%  
  select(species, decimalLatitude, decimalLongitude, locality, stateProvince, year)  
  
# 2.  
frogs_sf <- st_as_sf(frogs_clean, coords = c("decimalLongitude", "decimalLatitude"), crs = 4326)  
  
# 3.  
phl_provinces <- st_read("PHL_adm1.shp")
```

```
## Reading layer 'PHL_adm1' from data source 'C:\Users\ADMIN\Downloads\PHL_adm1.shp' using driver 'ESRI'
## Simple feature collection with 81 features and 9 fields
## Geometry type: MULTIPOLYGON
## Dimension:      XY
## Bounding box:   xmin: 116.9283 ymin: 4.58694 xmax: 126.6053 ymax: 21.07014
## Geodetic CRS:   WGS 84
```

```
phl_provinces <- st_make_valid(phl_provinces)
relevant_provinces <- phl_provinces[frogs_sf, ]

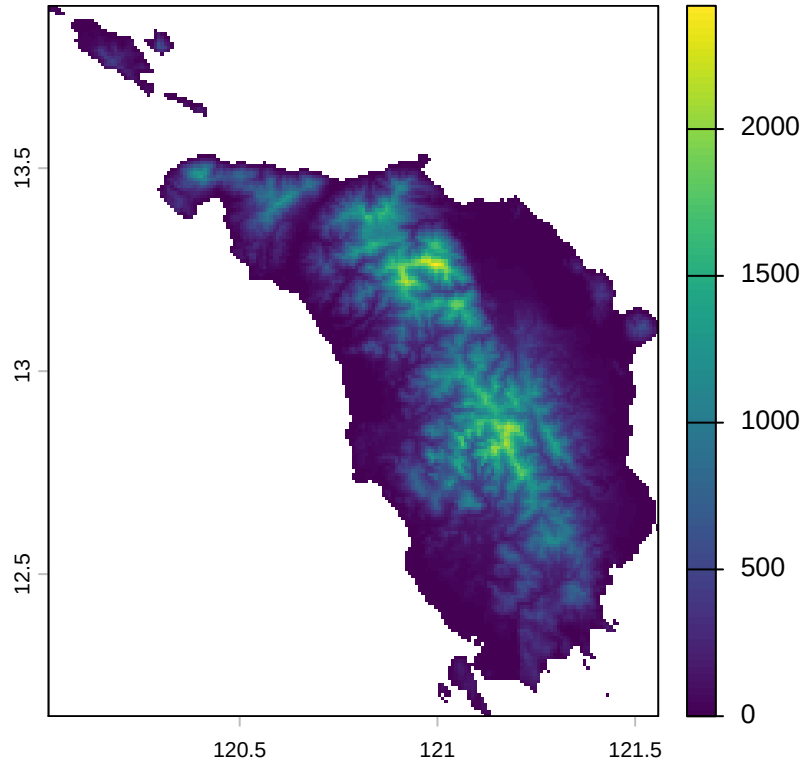
plot(st_geometry(relevant_provinces), main = "Relevant Provinces for Species Distribution")
```

Relevant Provinces for Species Distribution



```
dem <- rast("PHL_msk_alt.tif")
dem_crop <- crop(dem, relevant_provinces)
dem_clipped <- mask(dem_crop, relevant_provinces)
plot(dem_clipped, main = "Clipped Elevation Model for Study Area")
```

Clipped Elevation Model for Study Area



```
# 4
frogs_sf <- st_transform(frogs_sf, 4326)
relevant_provinces <- st_transform(relevant_provinces, 4326)
dem_df <- as.data.frame(dem_clipped, xy = TRUE)
colnames(dem_df) <- c("x", "y", "elevation")
ggplot() +
  # Layer 1: dem
  geom_raster(data = dem_df, aes(x = x, y = y, fill = elevation)) +
  scale_fill_gradientn(colors = terrain.colors(10), name = "Elevation (m)") +

  # Layer 2: Province
  geom_sf(data = relevant_provinces, fill = NA, color = "black", linewidth = 0.5) +

  # Layer 3: Species Points
  geom_sf(data = frogs_sf, aes(color = species), size = 2, alpha = 0.9) +

  theme_minimal() +
  labs(title = "Species Geographic Distribution",
       subtitle = "Overlaid on Provincial Boundaries and Elevation",
       x = "Longitude", y = "Latitude") +
  theme(legend.position = "right")
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using 'geom_tile()' instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using 'geom_tile()' instead.
```

Species Geographic Distribution Overlaid on Provincial Boundaries and Elevation

